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How green social network affordances enhance pro-environmental behaviour?

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Abstract

Though social media is considered to enhance environmentally conscious behaviour, there has not been sufficient investigation on the effects of specific affordances pertaining to green social networks (GSN). Also, the mechanism by which these affordances affect pro-environmental behaviour (PEB) has not been explored in depth. Addressing these research gaps, this study first introduces three context-specific GSN affordances, such as green self-identification, green broadcasting, and green social endorsement. Further, drawing inspiration from social influence theory and the Stimulus-Organism-Response framework, the study put forth a novel framework that explains the varying effects of the conceptualized GSN affordances on PEB through the mediating mechanism of collective intention. The framework also indicates perceived informational and emotional values as moderators that accentuate the indirect influence of GSN affordances on PEB. Adopting a time-lagged multi-wave online survey design, 314 valid responses were obtained. The results of the moderated mediation analysis reveal that: (1) collective intention partially mediates the relationships between green broadcasting and PEB, and between green social endorsement and PEB; (2) collective intention fully mediates the relationship between green self-identification and PEB. Further, the indirect effect of green self-identification on PEB is positively moderated by emotional values and not moderated by informational values. Interestingly, the indirect effects of green broadcasting and green social endorsement on PEB are positively moderated by both informational and emotional values. These findings offer significant implications for researchers and practitioners who look out for online communities to promote actual PEB.

KEYWORDS

affordances, green behaviour, green social networks, pro-environmental behaviour, social media, sustainability

1 | INTRODUCTION

In light of the looming environmental issues, the UN 2030 Agenda for Sustainable Development reflects a plan of action to protect people and the planet (https://sdgs.un.org/2030agenda). While several such bodies highlight the detrimental environmental impacts caused by business entities, the latest IPCC report claims that 'human-induced climate change' is causing widespread disruption to the planet (Palanivelu, 2022). Therefore, it is necessary to focus on mechanisms that induce individual pro-environmental behaviour (PEB). This forms the motivation for the current study to propose a novel framework that provides insightful directions to enhance PEB among individuals.

PEB emphasizes individuals' responsible actions that either promote sustainable usage of natural resources or reduce negative environmental consequences (Keizer & Schultz, 2018; Lange & Dewitte, 2019; Wesselink et al., 2017). Some of the examples include the adoption of energy-efficient transportation and sustainable waste management at home (Bissing-Olson et al., 2016; Wesselink et al., 2017). Research studies have consistently considered several motivational factors for PEB. Some of them include environmental awareness (Fernandez et al., 2017; Maidunoma & Falmatami, 2018; Vicente-Molina et al., 2013), social norms and influence (Culiberg & Elgaaied-Gambier, 2016; Keizer & Schultz, 2018; Xiao et al., 2023), and environmental emotions (e.g., pride and empathy; Coelho et al., 2017; Manika et al., 2021; Wang, Sheng, et al., 2023). However, it is still unclear how specific external mechanisms can induce the above-mentioned motivational factors (Alzubaidi et al., 2021; Han et al., 2018; Huang, 2016). Also, theoretically and empirically sound research pertaining to the enhanced understanding of PEB drivers is scarce (Xiao et al., 2023).

In this regard, social interactions and influence are generally considered external stimuli for PEB (Han et al., 2018; Huang, 2016; Li & Wu, 2020; Pittman & Abell, 2021; Salim et al., 2020; Xiao et al., 2023). Interestingly, social media has become one of the essential social interaction tools in the current digital era. Indeed, a statistics report showing 4.76 billion social media users clearly depicts the mounting increase in social media usage (Chaffey, 2023). Particularly, netizens are shifting towards specific online communities[1] to engage in context-specific activities (Chang et al., 2023; Fernandez et al., 2016; Han et al., 2018; Klassen et al., 2021; Wang, Liu, et al., 2023; Zhu et al., 2020). These specific online communities can be more effective than generic social media because they drive homophily and provide relevant and focused content, thereby ensuring utility (Andrews, 2002; Hajli, 2018; Wang et al., 2022). Therefore, given the significance of specific online communities, recent research studies have started to explore the effect of green online communities, also referred to as green social networks (GSN),[2] on green practices such as eco-purchasing (Byrum, 2019; Luo et al., 2020; Nekmahmud et al., 2022), green voice behaviour (Bhatti et al., 2020), and green collaborative practices (Salim et al., 2020; Shen et al., 2023). Also, research demonstrates that online relationships may induce PEBs such as collaborative consumption (Małecka et al., 2022). These arguments make GSN a worthwhile investigation in PEB research. Said that, the investigation regarding how specific online communities afford individuals the opportunity to indulge in PEB is unclear and vague. In this regard, the novelty of the current study lies in the introduction of context-specific GSN affordances while investigating the GSN–PEB relationship.

Affordances, in general, are interpreted as action possibilities offered by a system (Leidner et al., 2018; Treem & Leonardi, 2013). Specifically, social network affordances are found to enhance several prosocial and collective behaviours (Ahuja et al., 2018; Kelly et al., 2022; Leidner et al., 2018; Sun et al., 2019). Though PEB is also a form of prosocial behaviour, the investigation of the GSN's effectiveness on PEB using the affordance lens is relatively sparse (Han et al., 2018; Lange & Dewitte, 2019). Addressing this call, the current research captures the significance of GSN affordances in PEB scholarship.

Further, as per prior research, the social influence theory (SIT) holds great relevance in studying environmentally conscious behaviour (Goldsmith & Goldsmith, 2011). According to this theory, community members' behaviours influence individuals' behaviours as the interactions help individuals re-evaluate their intentions (Venkatesh & Brown, 2001). In coherence, the current research employs the SIT as a fundamental basis to further understand the proposed linkage between GSN affordances and PEB.

The current research also contributes to the GSN and PEB literature in multiple ways. First, the study renders a novel contribution to the existing PEB scholarship by introducing and investigating the varying effects of specific GSN affordances, such as green self-identification, green broadcasting, and green social endorsement, on PEB. The introduced affordances possess unique characteristics, including their non-competitive nature, emphasis on both active and passive actors, and provision of both individualistic and social perspectives. Henceforth, they are expected to render a granular approach and provide a holistic understanding of GSN affordances in the PEB realm, given the fact that there is a lack of literature that provides specificity while investigating the motivating factors of PEBs (Xiao et al., 2023).

Second, having known the significance of affordances in PEB literature, it is also critical to note that affordances are only *potential capabilities* for behaviours/actions, while actualization of affordances involves *specific desirable behaviours/actions* (Volkoff & Strong, 2017). Therefore, in this research context, GSN affordances must be realized to induce specific behaviours, that is, PEB. Interestingly, such realization happens through the formation of desirable internal states, such as intentions and attitudes (Leonardi, 2011; Seidel et al., 2013). From these vantage points, we can understand that GSN affordances should induce appropriate internal states to generate PEB. However, existing research lacks sufficient investigation in this regard. Addressing this gap, this study puts forth a novel framework that portrays collective intention as the mediating internal state that bridges the GSN affordances–PEB relationship. The mediational conceptualization is grounded in the Stimulus-Organism-Response (S-O-R) framework, which posits that external environmental aspects will act as stimuli and evoke internal emotional states (organism), which in turn induce certain actions (response) (Jacoby, 2002; Mehrabian & Russell, 1974). This study hypothesizes GSN affordances as stimuli, collective intention as an organism, and PEB as a response.

Third, the research makes a significant contribution to the GSN–PEB literature through the introduction of collective intention from a socio-psychological perspective. Researchers have increasingly recognized that pro-individualistic intentions alone may not be sufficient to drive PEBs (Wesselink et al., 2017). Interestingly, collective intention takes the form of a socio-psychological dimension that reflects the additive capacity of individual intentions to drive constructive behaviours (Kern & Moll, 2017). Specifically, collective intentions and identities are found to influence environment-related actions (Barth et al., 2021; Harland et al., 2007). Also, research has strongly deciphered that social networking helps individuals develop collective intentions, such as we-intentions (Chen et al., 2020; Shen et al., 2011), and such intentions may increase individuals' commitments towards pro-social behaviours (Feng et al., 2022). These arguments again suggest a strong affinity between GSN affordances, collective intention, and PEB, thus providing a meaningful investigation.

Finally, the research is one of the initial studies that proposes perceived informational and emotional values as moderators in the relationship between collective intentions and PEB, thus contributing to the intention–behaviour gap literature. Also, the introduction of the moderators provides an insightful explanation of why and how certain GSN affordances fail or succeed in promoting PEB. Further, given the existing mixed findings about social network effectiveness (Cooper et al., 2019; Shen et al., 2023), a deeper examination of the process by which GSN affordances enable PEB is required. Also, research suggests that users' engagement in PEBs is not always consistent with their PEB intentions and concerns (Carrington et al., 2010; Tseng, 2016). In relation to this, existing studies have called for further investigation into the contingent factors that may prevent or accelerate the conversion of intentions to PEB (Grimmer & Miles, 2017; Tian & Liu, 2022). Interestingly, previous research claims that the informational and emotional values offered by the network members play key roles in the social network's effectiveness (Wang, Lu, et al., 2021). Perceived informational value may motivate individuals to convert their intentions into behaviours (Yang et al., 2016). Moreover, research suggests that a lack of perceived informational value may increase the intention–behaviour gap, especially in the context of behaviours such as ethical consumption (Hassan et al., 2016) and local food consumption (Birch & Memery, 2020). Along similar lines, it is reported that positive emotional values may reduce the intention–behaviour gap (Bhattacherjee & Sanford, 2009; Sultan et al., 2020), especially in online communities (Wang, Lu, et al., 2021). Together with these pieces of evidence, this study integrates the SIT (Kelman, 1958; Kelman, 1974) and the S-O-R framework (Jacoby, 2002; Mehrabian & Russell, 1974) to develop a moderated mediation model that clarifies how the relationships between GSN affordances, collective intention, and PEB are contingent upon perceived informational and emotional values.

The rest of the research paper is structured as follows: Section 2 presents the theoretical background and hypotheses development of this study. Section 3 presents the research methodology. Section 4 provides a description of the validity and reliability assessments using AMOS and hypotheses testing using the Process macro. Section 5 puts forth a discussion on the results, and Section 6 presents the implications, limitations, and future directions. Finally, Section 7 presents the conclusion of the research work.

2 | THEORETICAL BACKGROUND AND HYPOTHESES DEVELOPMENT

2.1 | The social network affordance lens

Affordance is an action possibility, and it has been described as 'the potential for behaviours associated with achieving an immediate concrete outcome' (Volkoff & Strong, 2013, pp. 822–823). Specifically, information systems (IS) literature reports that rightly perceived affordances help individuals use the IS in a meaningful way (Faraj & Azad, 2012). Also, previous research demonstrates that the primary role of IS in environmental practices is to create specific green action possibilities, and such action possibilities help individuals alike indulge in environmentally conscious behaviours (Seidel et al., 2013).

Though affordances are perceptual, they are developed based on the specific goal orientations of the actors and the basic interaction between the actors and the system (Leidner et al., 2018; Treem & Leonardi, 2013). Particularly, information technology (IT) affordances reflect the focus of attention that translates the properties of the system into behaviours (Faik et al., 2020). As Gibson (1977) notes, 'what the object affords us is what we normally pay attention to' (p. 134). Therefore, attention, basic interactions, and relationships with the system/actors help individuals construct perceptions about the affordances (Faik et al., 2020). Further, these uniquely perceived affordances induce behavioural outcomes (Leonardi, 2011). Additionally, it is well-known that social network affordances provide cues that encourage people to engage in specific, affordance-relevant behaviours (Vaast et al., 2017). This indicates that an affordance lens is essential to clearly explaining how GSNs, which are exclusively designed for individuals with green ideologies, may effectively enhance PEB.

Going forward, it should be noted that affordances are different from affordance outcomes/actualization (Leidner et al., 2018). For instance, the influence of affordances on behaviours is referred to as affordance outcomes. Researchers have also demonstrated the influence of perceived affordances on behaviours such as knowledge sharing (Sun et al., 2019) and sustainable work practices (Seidel et al., 2013). Further, studies have also expanded this understanding by investigating the relationship between 'affordances, psychological outcomes (internal states), and behavioural outcomes' (Huang & Zhou, 2021; Suh & Wagner, 2017; Xu et al., 2022). On a similar front, this study also engages in the investigation of how perceived affordances result in an internal state (collective intention), leading to a behavioural outcome (PEB). This study argues that when individuals strongly identify and perceive the benefits of the GSN affordances, they will be able to develop a sense of collective intention. This may further promote the actualization of the affordances. Since the affordances reflect the environmental context, the research treats PEB as an affordance outcome.

2.2 | GSN affordances on PEB

2.2.1 | Green self-identification on PEB

Research has recognized that green self-identity is a prime driver of eco-friendly behaviours (Barbarossa & De Pelsmacker, 2016; Sharma et al., 2020; Whitmarsh & O'Neill, 2010; Zhao et al., 2023). Individuals who perceive themselves as green users are more likely to indulge in green actions (like recycling and consuming organic food; Becerra et al., 2023; Shaw & Shiu, 2003; Sparks & Shepherd, 1992). This is due to the fact that the stronger they identify themselves as green consumers, the more sensitive they become to their actions that may impact the environment, and the more they will change their habitual behaviours to reduce the harmful environmental impacts (Barbarossa & De Pelsmacker, 2016; Biel, 2017).

While several studies have demonstrated the relationship between self-identity and PEB (Confente et al., 2020; Hansmann et al., 2020; Sharma et al., 2020; Zhao et al., 2023), it should be noted that these studies have considered self-identity as a pre-existing psychological trait of an individual. Whereas this study considers the perceived affordance of green self-identification as a dynamic possibility that the GSN affords for the individuals to develop authentic green self-identity. From the affordance perspective (Leidner et al., 2018), green self-identification is conceptualized as the green action possibility that facilitates individuals to identify themselves as environmentally conscious persons and connect with environment-related activities (Confente et al., 2020). The variable reflects the dynamic nature of the green self-identity. This dynamic nature merits more attention as it affords individuals the opportunity to develop a desirable identity to further exhibit PEBs. Also, the conceptualization of green self-identification exhibited in the study differs from the concept of self-presentation or virtual self-identity, which heavily focusses on the symbolization and display of the virtual self (Wallace & Buil, 2023). For instance, the green self-identification introduced in this study focusses more on the internalization that depicts how individuals perceive themselves as environment-focussed. According to past research, 'the internalization dimension is a fairly reliable predictor of prosocial behaviour, the symbolization dimension is a less consistent predictor', as internalization helps individuals to be internally motivated to indulge in prosocial actions (Winterich et al., 2013).

With properties such as information exchange and the persistent availability of context-specific synchronous and asynchronous information, social networks can act as vehicles to construct self-identity towards specific behaviours (Cao et al., 2015; Wallace & Buil, 2023). Also, research claims that a strong sense of self-identification can be developed through network exposure (Kim, 2018; Wang, Van der Werff, et al., 2021). Social network research has consistently demonstrated self-identification as an affordance that enables desirable behaviours (Lin, Xu, & Wang, 2022; Lin, Zhu, et al., 2022; Wang & Chen, 2021). In this way, GSN may also enable green self-identification, as individuals will be able to psychologically connect with the messages shared in specific green online communities (Dickinson et al., 2017) and vertical networks[3] like Care2 (www.care2.com). Interactions in such networks will be a window for individuals to identify themselves with green ideologies (Papacharissi, 2013). While generic social networks may also facilitate identity manipulation and perverse social capital (Maghrabi et al., 2014), specific social networks can mitigate this issue to an extent as individuals voluntarily become members of like-minded groups and share common goals.

These properties of the specific social networks (GSN in this case) may enable individuals to easily develop context-specific self-identity (green self-identification in this case) without the necessity for identity manipulation.

Subsequently, the green self-identity derived from such social networks may encourage PEBs (Brick et al., 2017; Wallace & Buil, 2023). Therefore, this research hypothesizes that the green self-identification affordance of the GSNs may motivate individuals to engage in PEB.

H1. Green self-identification affordance enhances PEB.

2.2.2 | Green broadcasting on PEB

Social network features, such as status updates, hashtags, original posts, and tags, allow individuals to quickly disseminate content to a wider audience (Ellison & Vitak, 2015; Zhang & Jung, 2023). This enables knowledge mobilization and diffusion at a low cost. Synthesizing these capabilities, this study conceptualizes green broadcasting as an affordance that the GSNs allow individuals to share or receive environment-related videos, text, and images with or from all the network members.

In general, broadcasting in online communities plays a greater role in influencing network members' behaviours (Kim & Shen, 2020; Subramani & Rajagopalan, 2003). In particular, green broadcasting can significantly contribute to the creation of environmental knowledge (Maidunoma & Falmatami, 2018), which is a crucial motivating factor for PEB (Dhir et al., 2021; Do Paço et al., 2019; Lee, 2011). Therefore, green broadcasting can be counted as an essential vehicle to quickly and effectively promote PEB (Simeone & Scarpato, 2020). However, from a contradictory standpoint, research claims that green broadcasting on social networks may lead to scepticism due to a lack of perceived information credibility (Luo et al., 2020; Su et al., 2021). In this regard, it should be noted that the messages broadcasted by the members of a closed network may reduce scepticism and induce trust, which are essential for the adoption of green behaviour (Mezger et al., 2020). Another criticism of broadcasting is that it may not generate desirable outcomes when the broadcasted content is automated (Chen et al., 2015), and automated broadcasted messages are often treated as spam (Rao et al., 2021). However, members of exclusive online communities (GSN in this case) with a specific focus are less likely to indulge in social spam. Therefore, green broadcasting through such channels is expected to provide meaningful knowledge and induce PEB. Also, individuals are found to easily gravitate towards the messages that their peers broadcast. This gravitation may result in the successful adoption of environmentally conscious behaviour (Byrum, 2019; Lee, 2011). This argument highlights the significance of the current research, which emphasizes the information exchange between the users on GSN rather than focussing on generic social media communications.

Going forward, broadcasting affordance allows individuals to quickly disseminate information to members with strong as well as weak ties (Hemsley & Mason, 2012). Both strong and weak ties are beneficial for positive behavioural outcomes. For instance, strong ties facilitate trust and motivate individuals to easily share resources. Weak ties help individuals gather resources from heterogeneous people separated by temporal and social distances (Haythornthwaite, 2005). Since GSNs consist of both strong and weak ties, broadcasting environment-related content on such networks will allow individuals to take advantage of knowledge gained from both ties when they practice PEB.

Green broadcasting affordances may facilitate information accessibility and deliverability, thereby improving the likelihood of conversations (Pearson et al., 2016). Such continuous engagement may accentuate PEB. Said that, engagement can be both active (posting announcements) and passive (viewing the announcements made by other active members). Broadcasting may, therefore, benefit both active and passive actors. In fact, previous research also claims that even passive actors may benefit from broadcasting affordances through network exposure; they can subsequently acquire knowledge and indulge in productive behaviours (Lee & Cho, 2020). In sum, this research strongly affirms that the green broadcasting affordance that the GSNs afford may effectively enhance the PEBs of all the network members.

H2. Green broadcasting affordance enhances PEB.

2.2.3 | Green social endorsement on PEB

Endorsement, in general, democratizes the information and pluralizes the voices, thereby enhancing the credibility of the content and its influence on individuals (Castello et al., 2013). The social network features, such as emojis and other paralinguistic digital affordances, help individuals express their interests and emotions, thereby evoking social endorsement (Fernández et al., 2022; Thai & Wang, 2020). Research has demonstrated that features such as positive comments effectively signal social endorsement to network members (Lee & Eastin, 2021; Thai & Wang, 2020; Yu & Hu, 2020). In this light, the current study conceptualizes green social endorsement as an affordance that the GSN enables users to seek or provide instant validation and social support for the pro-environmental content shared in the network. Prior studies have also noted the existence of social endorsement affordance as the online social networks allow users to access and engage with the content by reading or signalling with likes and comments (Sun et al., 2021).

Further, more than the endorsements offered by famous personalities, social endorsements in GSN may have a stronger influence. This is due to the fact that individuals in general are more inclined towards social in-groups (Tajfel, 1982). The simple actions taken by the network members/peers are more organic in nature and therefore may result in favourable behavioural outcomes (Thai & Wang, 2020). Also, when individuals observe that their network members follow or like a particular piece of content, they may also have a positive affinity for the content. Social endorsements, such as the number of likes and comments, will be embedded in the individuals' browsing experience, thereby providing repeated exposure and attention to the message. This will act as a social cue to influence individuals to engage in target behaviours (Thai & Wang, 2020). For instance, a greater number of appreciations, recognitions, and reposts reflect the value and credibility of the content (Byrum, 2019; Thai & Wang, 2020). This way, group consensus can be easily achieved towards the content posted in the network, and such consensus may motivate individuals to react or behave in desirable ways (Oh et al., 2017; Thai & Wang, 2020). Similarly, from a green-based research perspective, social endorsement (through the number of likes or re-posts) is argued to be one of the important drivers of green behaviours (Gerpott & Mahmudova, 2010; Shen et al., 2023). In sum, when individuals receive positive endorsements for the green content, they may acquire the required social support (Wohn et al., 2016). This may serve as a strong intrinsic motivation to readily engage in PEB (Wesselink et al., 2017). This research therefore hypothesizes that,

H3. Green social endorsement affordance enhances PEB.

2.3 | Influence of proposed GSN affordances on collective intention

2.3.1 | Green self-identification on collective intention

Self-identity is often interpreted as an action potential offered by social networks, which stimulates specific social intentions and behaviours (Qiu et al., 2021; Wallace & Buil, 2023). In general, previous research suggests that different social groups with different purposes generate corresponding self-identities (Kawakami & Dion, 1993). Similarly, GSNs that act as specific social groups with a specific purpose may generate environmental self-identity (Wallace & Buil, 2023).

Social identity formed in social networks is claimed to be a salient driver of collective intentions (Bagozzi & Dholakia, 2006; Chen et al., 2020; Cheung & Lee, 2010; Shen et al., 2013). However, research also portrays that social identity can also be created to gain certain social rewards and avoid unfavourable outcomes (Becker & Tausch, 2015; Gable & Strachman, 2008). Therefore, social identity alone may not be sufficient to develop collective intentions. In similar lines, researchers suggest that social self-identification formed in a community/network can constructively generate oneness because individuals build self-identities based on the interactions they have in the network (Pan et al., 2017) and the social roles they perform (Van Dijk et al., 2015). Only when there is a strong sense of self-identification will there be a possibility for we-membership because interpersonal differences derived from self-identification are equally important to bring richness to the group (Gilbert, 2014). Self-identity has also been shown to foster social connectedness since it enables members of a community to recognize their uniqueness as well as their commonality (Christensen et al., 2004). Research therefore claims that strong self-conception develops relevant social identity (Zahavi, 2021), which is positively correlated with collective intention (Bagozzi & Dholakia, 2006). This indicates that self-identity can enhance collective intentions and actions (Chen et al., 2020; Van Stekelenburg, 2013). Besides, self-identity promotes relational self and therefore connects individuals with the group when the conception of self and vision of the group are aligned (Kark & Shamir, 2002). In congruence with this, this research expects that green self-identification may help individuals relate themselves to the group and facilitate collective intention as the group's and individuals' identities are aligned with green ideologies. Existing research has also demonstrated a positive link between green self-identity and pro-environmental intentions (Ashraf & Musarskaya, 2018; Lalot et al., 2019). This research extends the existing literature by highlighting the potential link between green self-identification and collective intention towards PEB.

H4. Green self-identification affordance enhances collective intention towards PEB.

2.3.2 | Green broadcasting on collective intention

Researchers note that broadcasting affordances in social networks can serve as a tool to develop connectedness in the network (Hu & Chaudhry, 2020; Vitak, 2012). In particular, specific social networks evoke sufficient critical mass, thereby subsequently developing a sense of community and collective intentions (Liao et al., 2020; Shen et al., 2013). Therefore, users prefer specific social networks to broadcast altruistic messages (e.g., pro-environmental actions) through videos, messages, and pictures (Han et al., 2018).

It is also observed that technologies and strategies that support PEB alone may not actualize desirable behaviours (Klöckner, 2015). For example, electric car users may perceive that they are environmental benefactors and comfortably use their cars beyond the actual usage requirements (Klöckner et al., 2013). This ultimately deteriorates the purpose of their purchase. Therefore, the inducement of collective responsibility towards the environment requires effective persuasive communication that enhances sufficient awareness about one's actions. Interestingly, specific social networks, such as GSN, can facilitate collective responsibilities as they allow the broadcasting of related and significant messages to a large and relevant group of stakeholders (Ghali et al., 2016; Klöckner, 2015; Sun & Xing, 2022). Further, environmental awareness (e.g., awareness of the environmental objectives, environmental consequences of one's behaviour, and biospheric values) can be easily acquired through the green broadcasting affordance of the GSNs. This awareness helps individuals build a sense of community, which is an essential element of collective intention (Francis et al., 2012). Therefore, this study proposes that broadcasting environment-related messages through GSNs may help network members develop collective intentions.

H5. Green broadcasting affordance enhances collective intention towards PEB.

2.3.3 | Green social endorsement on collective intention

The achievement of social reputation can significantly enhance individuals' green intentions (Groening et al., 2018). Appreciation cues in online communities facilitate recognition for a piece of shared content, and this recognition provides a sense of social endorsement (Lee et al., 2021). Such social endorsement not only provides social support, but it also reflects trust, credibility, and consensus (Borah & Xiao, 2018). The content of the endorsed message is perceived to have higher integrity (Kim, 2018). This trust and credibility may enhance collective intention (Li et al., 2022) and behaviours related to environmental concerns (Gerpott & Mahmudova, 2010). Also, when users receive such digital recognition in interactive IS, they tend to develop positive relationships with the network members, and this form of relationship is essential to developing collective commitment (Shen et al., 2011). Research strongly affirms that recognition drives social influence and leads to a sense of collective feeling like social bonding while using IS applications (Hernandez et al., 2011), particularly in specific online communities (Nivedhitha & Sheik Manzoor, 2020). Further, social endorsement strongly enhances positive emotional experiences (Wang, Thai, et al., 2021), and interestingly, we-intentions are developed when individuals experience positive emotions (Morschheuser et al., 2017). Considering these arguments, this research anticipates that green social endorsements via digital recognition features offered by GSNs induce collective intentions towards PEB among GSN users.

H6. Green social endorsement affordance enhances collective intention towards PEB.

2.4 | Influence of collective intention on PEB

According to Tuomela (2008), 'collective intentionality in its central sense is based on "we-thinking," thinking in terms of a "we-perspective"'. (p. 3). On similar lines, this research represents collective intention as an individual's perception of the extent to which he/she, along with the group (members in the GSN), intends to participate in actions related to environmental conservation.

Collective intentions represent how individuals assume common goals and tend to act in unity (Tuomela, 2000; Tuomela, 2005). They reflect shared volition, which may encourage individuals to engage in pro-social behaviours. Also, the outcome of collective intentions is essentially social in nature, and collective intentions can drive both group and individual behaviours (Gilbert, 1989). On the other hand, even though PEBs are referred to as individual behaviours, they predominantly include pro-social actions, and moreover, individuals engaging in such behaviours have common green goals (Steg et al., 2014; Unsworth et al., 2013). In sum, PEB can be represented as a form of social action arising out of environmental concern (Amoah & Addoah, 2021). Therefore, collective intention offers an appropriate and new space to study PEB.

Previous research has explored the effect of collective intention on collective behaviours such as participation in multiplayer games (Li & Suh, 2021), group communication (Shen et al., 2013), and socially responsible behaviours (Chen et al., 2020). From a pro-environmental research perspective, it is reported that GSNs help individuals develop a shared commitment and such a shared commitment is critical to PEB (Ghali et al., 2016; Huang & Zhou, 2021). Previous studies have also claimed that a sense of community promotes PEBs (Rees & Bamberg, 2014). Also, research argues that to effectively transform individuals' intentions into behaviours, they must believe that their peers also have similar intentions (Shen et al., 2011). This research therefore contends that individuals will be more likely to engage in green-related behaviours when they perceive that the members of the GSN also intend to engage in green-related behaviours.

Further, SIT describes that individuals internalize social influence when they perceive that members have shared goals and commitment; such perception helps them to engage in socially induced behaviours (Davis et al., 1989). In connection to this, we can see that social influence accentuates intrinsic motivation to indulge in green behaviours (Van Der Linden, 2015), for example, green electronic product purchases (Ali et al., 2020). Based on the tenets of the theory, this research contemplates that collective intention will act as a social influence to promote individual PEBs.

Interestingly, individuals' intentions and behaviours are derivatives of collective intention (Searle, 1990). For example, 'I do' is a derivative of 'We intend to do, or we all do' (Searle, 1990). Searle (1990) explains this using football as an example. When a football team attempts to execute a pass play, members will have a collective intention, that is, to execute a pass play. Interestingly, this collective intention will also drive individual behaviours as each player will indulge in their individual assignments and make a specific contribution (e.g., 'I am blocking the defensive end'). This explains that collective intention may strongly induce individual behaviours too. In addition to this, studies suggest that these group referent intentions should be taken into account when investigating the behavioural outcomes derived from social network usage (Chen et al., 2015; Shen et al., 2013). These interesting claims motivate the current research to further understand PEB through collective intention. When the members of the GSN perceive that the entire group intends to indulge in PEB, they will be influenced by the social energy. Henceforth, they attempt to make a specific individual contribution, such as energy conservation and the purchase of sustainable products.

Further, the transition to a sustainable and low-carbon world requires more individualistic contributions than institutional interventions. The formation of ActNow, the UN campaign, resonates with the significance of individual actions such as driving an electric vehicle and taking public transport (https://www.un.org/en/actnow). Individual action is considered the core element of climate responsibility (https://www.niti.gov.in/life) because these individual actions will collectively promote positive environmental effects. Therefore, this study focusses on individuals' contributions to the environment and proposes collective intention as a core motivational resource for such individual contributions.

H7. Collective intentions enhance PEB.

2.5 | Collective intention as a mediator

The SIT states that social influence caused by a particular community affects one's intentions and behaviours, especially when individuals accept and internalize the social influence (Kelman, 1958). Literature on IS applications and social networks has continuously demonstrated the significance of social influence in facilitating collective intention (Chen et al., 2020; Shen et al., 2011) and PEB (Han et al., 2018). However, the S-O-R framework is essential to further shed light on how social influence through social networks enhances PEB since the framework focusses on how particular social cues facilitate internal processing and lead to specific behaviours. Therefore, in this research context, the S-O-R framework facilitates a deeper investigation into the relationships explained by SIT.

The S-O-R framework is extensively used to explain individual behaviours, including specific consumer behaviours and green practices (Hameed et al., 2022; Tang et al., 2019). The three stages in the S-O-R framework include stimulus, organism, and response. Stimulus (S) is either an external or internal cue that an individual encounters at a point in time. These cues will affect the internal states of the individuals (O), which in turn will lead to specific responses/behaviours (R) (Mehrabian & Russell, 1974). Therefore, the organism (O) is referred to as the first-order outcome of the stimulus, and it intervenes between stimulus and response (Mehrabian & Russell, 1974). Further, literature on IS affordances explains that affordances stimulate psychological and behavioural outcomes (Leidner et al., 2018; Volkoff & Strong, 2013; Xu et al., 2022). Also, collective intention is often referred to as one's internal state (Shen et al., 2013). Equating the S-O-R framework with the above-mentioned evidence, the present research identifies GSN affordances as stimuli and expects that they induce collective intention, ultimately leading to desirable behaviours.

First, self-identity emerges through social cues (Rise et al., 2010). Therefore, green self-identification via social networks can be considered a potential affordance as GSNs generate rich, green-related social cues (Khare & Kautish, 2021; Wise et al., 2010). Green self-identification can be self-rewarding and intrinsically motivating (Chi, 2021). Unarguably, intrinsic motivation is positively correlated with collective intention (Riar et al., 2023; Zhang et al., 2017). When individuals identify themselves with specific goals and actions, they will actively engage in community participation to express their identity (Floridi, 2011; Shen & Khalifa, 2014). Applying this, we can argue that when individuals identify with environmental consciousness, they tend to become more active in GSNs, which is essential to develop collective intention towards environmental actions. Next, broadcasting on social networks helps members develop group awareness, which positively affects we-intention (Maidunoma & Falmatami, 2018; Shen et al., 2011). Even marginal PEBs of individuals, such as conservation of water at home, can be broadcast to the entire network, and the effect of such content will be even more influential when it hits the right audience (exclusive networks for sharing environmental activities; Hynes & Wilson, 2016). Therefore, a GSN that affords the broadcasting of environment-related content to a desirable population can create positive synergy and develop collective intention among the members. Finally, recognition and social support are essential to develop shared commitment among social network users (Yang et al., 2016). This creates social value for further cooperative actions (Liao et al., 2020). Therefore, social endorsement through likes and comments may foster emotional attachments (Borah & Xiao, 2018) and shared aspirations (Wang, Thai, et al., 2021). This supports the proposal that social endorsement in GSNs may enhance collective intention towards pro-environmental actions.

The above-mentioned relationships exhibit a stimulus–organism sequence.

Further, to actualize the perceived affordances to behavioural outcomes, intention formation is critical. This research signifies that collective intention is essential for enhancing PEB because PEB is a responsible act, considering the earth from a 'We-perspective' and not from an 'I-Perspective'. Therefore, the collective intention is expected to significantly exert an influence on PEB since it is regarded as the individuals' intention to indulge in pro-social actions while perceiving that all the members also have such constructive intentions (Bagozzi & Lee, 2002; Shen et al., 2013). Further, a statement illustrating that 'it is not worth me making the effort if everyone else does not' (Lorenzoni et al., 2007) perfectly explains the importance of collective intention to develop PEBs. It should also be noted that social oneness is integral to the creation of sustainable behaviours (Xing & Starik, 2017). Considering the fact that collective intention focuses on oneness, this research considers PEB as a response derived from collective intention.

The above-mentioned arguments explain the organism–response sequence.

In line with the previous studies and the principles of the S-O-R framework, the study contends that GSN affordances function as external stimuli and promote desirable responses, that is, PEBs. Further, an organism in the S-O-R framework can be a perceptual or intermediary action that fundamentally reflects the affective and cognitive processes of individuals (Bagozzi, 1986; Jacoby, 2002; Kamboj et al., 2018; Loureiro & Ribeiro, 2011). The affective component focusses on the individuals' feelings triggered by environmental cues, and the cognitive component involves perceptual evaluation of the environment (Jacoby, 2002). Interestingly, collective intention reflects affective commitment towards a particular action (Shen et al., 2013). Collective intention also illustrates the cognitive evaluations of the individuals in the system (Zaibert, 2013). Therefore, this research treats collective intention as an organism that directs the stimuli to a response. Further, a few researchers have emphasized the socio-psychological perspective while narrating the S-O-R framework (Li et al., 2021). For instance, Chiu et al. (2019) demonstrated common bond attachments and common identity as organisms that intervene between virtual community cues (stimuli) and virtual community citizenship behaviours (response). Similarly, collective intention reflects a socio-psychological perspective as it is a combination of individuals' participatory intentions along with their perceptions of others' participatory intentions (Rachar, 2021). In coherence with previous research studies, the current research extends the understanding of the theory by adopting a socio-psychological perspective on the organism concept. In sum, the study contends that individual behaviours can be triggered by external stimuli through the socio-psychological internal state of the individuals.

H8a. Green self-identification enhances collective intention, which, in turn, enhances PEB.

H8b. Green broadcasting enhances collective intention, which, in turn, enhances PEB.

H8c. Green social endorsement enhances collective intention, which, in turn, enhances PEB.

2.6 | Moderating role of informational and emotional values

In many instances, pro-environmental intentions may not necessarily translate into desirable behaviour (Finisterra do Paço & Raposo, 2010; Mostafa, 2009). This may be due to the fact that online activities may generate slacktivism, that is, involvement in activities that require less effort (Bower, 2014). Particularly, research has also claimed that collective intention may not be sufficient to build moral responsibility and behaviours due to insufficient conditions for its transformation into behaviours (Searle, 1990; Smiley, 2017; Zaibert, 2003). Specifically, the intention–PEB gap was observed in previous research (Grimmer & Miles, 2017). Therefore, there is an emergent need to investigate the contingent factors that affect the conversion of collective intention derived from GSN affordances into real behaviours. Considering the fact that perceived values, in general, affect the relationship between intention and behaviours (Mazis et al., 1975), this study proposes that collective intention will increasingly enhance PEB, especially when individuals perceive high informational and emotional values.

Perceived informational value can be referred to as the cognitive element of the organism (Jacoby, 2002) because it refers to the individual's perceived sense of utility towards the information sought from the network (Sweeney & Soutar, 2001). Perceived emotional value refers to the individual's sense of positive affect and feeling generated by the network (Kaur et al., 2018; Khan & Mohsin, 2017). Therefore, the perceived emotional value can be associated with the affective dimension of the organism (Jacoby, 2002). Together, informational value acts as a cognitive mechanism, and emotional value acts as an affective mechanism. Also, both interact with another organism (collective intention) to deliver specific responses. Further, the S-O-R framework postulates that the typical sequence of stimulus to organism to response can also be interdependent (Jacoby, 2002). Also, Jacoby (2002) emphasizes that stimuli, organisms, and responses are fluid in nature, that is, there can be interactions between stimuli and organisms and between two organisms to induce a specific response. Also, as defined in the Venn diagram denoting the S-O-R framework (Jacoby, 2002), the concept of 'consciousness' is treated as a stimulus, but at the same time, it is also represented as an organism representing both affect and cognition (Jacoby, 2002). Following these guidelines, this study treats informational value (a form of cognition) and emotional value (a form of affect) as part of consciousness. Then the study contends that they interact with another organism (collective intention, which comprises both affective and cognitive components) to deliver the response (PEB).

From a PEB perspective, individuals may gain collective intention via specific GSN affordances. However, in order to transform those intentions into behavioural actions, individuals require valuable information about the finer details and procedures involved in pro-environmental actions. For example, the rate of recycling was found to be high when participants were exposed to sufficient recycling-related information (Iyer & Kashyap, 2007; Pichert & Katsikopoulos, 2008). Additionally, according to the green literature, when individuals perceive that the information related to environmental concerns is of value, they will clearly understand their role in the environmental consequences and have better knowledge about the whole process of environmental actions (Abrahamse, 2019). This informational value reflects the utilitarian aspect (Li, 2013), which is essential for individuals' intended actions (Sánchez-Fernández & Jiménez-Castillo, 2021). Congruently, high perceived informational value has been substantiated to fuel proactive behaviours in individuals. With valued information, individuals can invest their intellectual and/or material resources in an appropriate manner towards the targeted end (Wang, Lu, et al., 2021). In GSNs, informational value will lend rational approval (Xing et al., 2018; Zha et al., 2018) to their environmental pursuits, thereby consolidating their commitment to the network and its objectives. Informational support offered by valued information opens up avenues for network users to indulge in experiments and real-life actions (Wang, Lu, et al., 2021). Of late, scholars have propounded that informational value in social networks is a route to value co-creation extending beyond the business-consumer ambit, such that collaborative intentions are translated into individual tangible actions (Gebauer et al., 2013; Liu et al., 2020; Nadeem et al., 2020). Also, given that GSN users are sensitive to pro-social/pro-environmental information (Wang et al., 2022), valued information in GSNs can trigger constructive dialogue in the collective. It can thus accentuate the capability of collective intentions to drive subsequent appropriate actions. Endorsing this, social network studies advocate that valued information (in the form of advice or recommendations based on experience or expertise) serves as a reference point for social network users to follow the content (Wang, Lu, et al., 2021; Zha et al., 2018). It further increases their self-efficacy in terms of the success of their subsequent actions (Lin et al., 2015). That is, there is a high likelihood that their collective intentions actualize into PEBs when they believe that their engagement in certain PEBs will lead to the achievement of the intended outcomes. Besides, the SIT in general states that the value of the information received from society will allow individuals to undergo desirable behavioural changes. That is because the more the individual places value on the information, the more confident one becomes with regards to one's attitude towards something; this then guides individual behaviour (Spreng & Page Jr, 2001). This implies that, in the current context, perceived value will strengthen the individual's belief in collective intention, thereby easing the transition of these intentions into PEB. In other words, when individuals perceive high informational value, they tend to see greater value in the community and intend to contribute their share to it (Yang et al., 2016). Therefore, for individuals with higher levels of perceived informational value, collective intention (organism) will be more convincing in eliciting PEB (response) in comparison to individuals with low perceived informational value.

Besides the above, this study claims that emotional values as perceived by individuals may strengthen the influence of collective intention on their PEB. Prior research has shown that affective values (e.g., emotions like happiness, pride, and pleasure) play a role in bridging the intention–behaviour gap (Bhattacherjee & Sanford, 2009; Mohiyeddini et al., 2009). This indicates that positive emotions unconsciously motivate individuals to transform their intentions into actions (Chen & Bargh, 1997). In line with this, emotional values are surmised to play a role in green behaviour (Gonçalves et al., 2016; Khan & Mohsin, 2017; Lin & Huang, 2012). Especially emotional values may help individuals enhance the transformation of green intentions into pro-environmental actions (Joshi et al., 2021). Emotional value reflects the intrinsic reward/perception of positive affect (Winterich & Barone, 2011). Individuals with such positive emotions may exhibit stronger we-intentions in IS-based communities (Zhao et al., 2017), thereby enhancing their engagement in PEB.

Driven by these sets of evidence, this research hypothesizes that:

> **H9.** Perceived informational value moderates the relationship between collective intention and PEB such that the mediating effect of collective intention on the relation between GSN affordances (green self-identification [H9a], green broadcasting [H9b], and green social endorsement [H9c]) and PEB is strengthened when the individuals perceive higher levels of informational value.

> **H10.** Perceived emotional value moderates the relationship between collective intention and PEB such that the mediating effect of collective intention on the relation between GSN affordances (green self-identification [H10a], green broadcasting [H10b], and green social endorsement [H10c]) and PEB is strengthened when the individuals perceive higher levels of emotional value.

The hypothesized model, depicted in Figure 1, illustrates the relationships between the research variables.

3 | RESEARCH METHODOLOGY

The study confirmed the specificity of the social networks under investigation by ensuring that the participants' responses were based on their experiences with specific forms of social networks, that is, GSN in this case. Since there is a debate about information overload and irrelevance in generic social media (Guo et al., 2020), individuals have started to prefer exclusive social networks or online communities on general social media. The usage of specific GSNs for environmental behaviour has also become prevalent in recent times (Fernandez et al., 2016; Nekmahmud et al., 2022). GSNs can either be specific communities on generic social media platforms or exclusive platforms that facilitate environmental dialogue. Examples include Treehugger, Care2, and @Greenpeace International on Facebook. Following this, the present research identified sample sources. They include specific online communities on generic social media

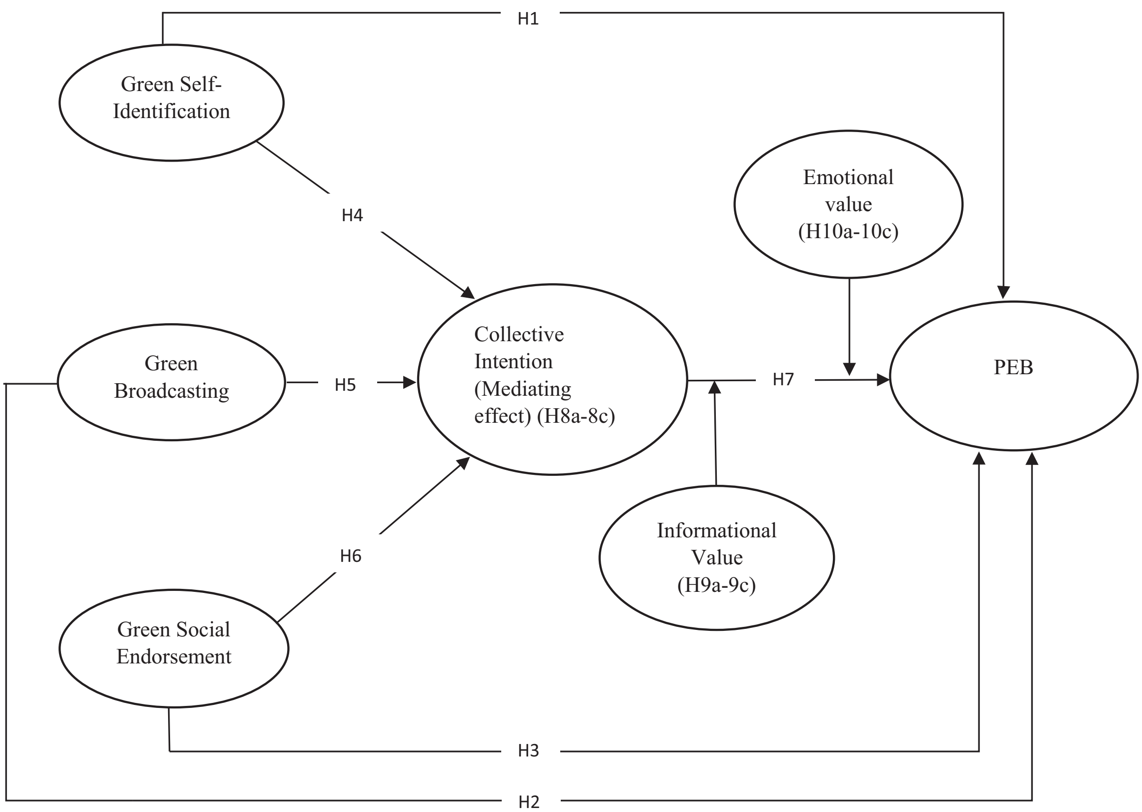

FIGURE 1 Hypothesized model.

(e.g., @igbconline on Facebook) and exclusive social networks related to PEB (e.g., Greenforum.com). To further facilitate the generalizability of the study, this research included respondents who follow/use all forms of GSNs. The generalizability across these social networks can also be observed in previous environment-related research works (Hynes & Wilson, 2016; Shah et al., 2021; Shen et al., 2023). Generalizability is primarily driven by the GSN commonalities with respect to the purpose of formation, type of interactions, and functional features of online networks (such as accessibility to relevant information, network exposure, and the ability to post, comment, and share). Also, these social networks help reach a relevant audience, develop a decentralized network, impart sustainability-related education, attract users' attention towards PEB issues, and motivate stakeholders to engage in several PEBs (Zhang et al., 2022).

Using the purposive sampling method, the study gathered appropriate samples. Further, the questionnaire was also sent via the survey market tool, SurveyMonkey, with a qualifying question (Are you a member of any online community that focusses on environmental consciousness?) to validate the appropriateness of the sample before allowing them to fill out the questionnaire.

An explicit explanation of the research objectives and concept of PEB was given along with the questionnaire. A total of 412 appropriate users volunteered to participate in the survey.

Since the current research involves a survey method, it is susceptible to several biases that may affect the results. Therefore, several preventive mechanisms, such as temporal separation of variables and anonymity during the administration of questionnaires, were established to minimize the bias (Podsakoff et al., 2003). Detection tests such as Harman's one-factor and Common latent factor analyses were performed to ensure that the study results were not affected by the bias-related concerns (Podsakoff et al., 2003).

A time-lagged multi-wave survey design facilitates temporal separation between the measurement variables. Such data collection is used to reduce bias-related issues such as social desirability and common method bias, and the method reduces concerns related to causality in cross-sectional designs as well (Podsakoff et al., 2003). For these reasons, several studies in the past have employed a three-wave survey design while investigating the mediating effects (Bednall & Sanders, 2017; Pasricha et al., 2023). Following the guidelines, the study employed a three-wave design by collecting responses with a 3-week time lag between every wave. Participants were requested to complete three online surveys at different time intervals. Though the responses were matched based on the unique identifier (corresponding to an email ID) across the waves, the identification was kept confidential and destroyed at the end of the research. Thus, confidentiality and anonymity were ensured.

First, independent variables, that is, green self-identification, green broadcasting, and green social endorsement, were measured at time 1. Out of 412 respondents, only 400 responses were received, and finally, 393 responses were selected after discarding the invalid

TABLE 1 Descriptive statistics.

Characteristics	Levels	Frequency	Percentage
Gender	Female	186	59
	Male	128	41
Age	21–30	170	54
	31–45	101	32
	>45	43	14
Education	UG	121	38
	PG	124	39
	Doctorates	69	23
Topics that interest you in green social networks	Campaigns	93	30
	NEWS	90	29
	Fund-raising information for environment related cause	75	24
	Information about eco-friendly products before purchase	56	17

responses. Then mediating and moderating variables, that is, collective intention, emotional values, and informational values, were measured at time 2. 375 responses were received. Finally, the dependent variable, that is, PEB, was measured at time 3. The received responses were reduced from 375 to 334, and further, the invalid and missing responses were discarded. Finally, 314 valid responses were retained for analysis. A *t*-test was performed to find if there was any pattern between the demographic data and the missing responses. The results did not show any significant patterns.

As illustrated in Table 1, 59% of the sample was male. In terms of age, the majority of the individuals were in the range of 21–30 years (54%). As regards education, 23% of the respondents were PhD holders, 38% had undergraduate degrees, and 39% had postgraduate degrees. Among the topics that interest individuals in GSNs, campaigns fetch the first position, with 30% of the respondents claiming their preference, followed by news with 29%, fundraising information with 24%, and information about eco-friendly products with 17%. The demographic details are presented in Table 1.

3.1 | Measures

Existing scales were utilized by either adopting or adapting them to suit the context of the current research. All the scales were displayed on a 5-point Likert scale from 'strongly disagree' to 'strongly agree'. To validate the survey instrument, both expert opinions and the pilot study were undertaken. With the help of five experts, consisting of professors and young researchers with an interest in PEB research, the structured questionnaire was fine-tuned in terms of appropriateness and readability. Further, the pilot study was conducted among a small group of individuals who were specifically members of online communities pertaining to environment-related activities. Upon suggestions, certain wordings of the questionnaire were changed to enhance understandability.

Green self-identification was measured using items borrowed from Harrigan et al. (2018). The scale focuses on how the GSNs stimulate individuals to identify themselves as environmentally responsible individuals (Harrigan et al., 2018; Jeannerod & Pacherie, 2004). Green broadcasting was measured using items adapted from the creation dimension of customer engagement on social networks (Thai & Wang, 2020). The creation component scale (Thai & Wang, 2020) focussed on how individuals can post and share content on social networks. Since the content, once posted, is obviously broadcast to the network, the scale is found to be appropriate to measure broadcasting. In this research context, the green broadcasting affordance scale measures the potentiality of GSN in allowing individuals to spread environment-related content to a large set of targeted audiences within a short span of time. Green social endorsement was measured using items adapted from Gan (2017) and Thai and Wang (2020) The social endorsement affordance scale measures the extent to which GSNs allow the aggregation of the users' recognition for content (shared by peers) available in the network. The aggregation can be brought about through single-click affordances such as positive emojis (e.g., likes, upvotes, and smileys) and appreciation in the form of text, comments, and other cues. Collective intention was measured using items adapted from Iyer et al. (2007) and Shen et al. (2013).

Perceived emotional value was measured using the items from Sweeney and Soutar (2001). The scale measures the extent to which individuals perceive the affective value, such as emotional gratification and aroused feelings, generated from the content shared in GSNs (Joshi et al., 2021; Sweeney & Soutar, 2001). Perceived informational value was measured using the items from Okleshen and Grossbart (1998) and Yang et al. (2016). Emotional value measures the extent to which individuals perceive the functional value of content such as knowledge acquisition and the relevance of the information sought from the GSNs (Wan et al., 2017; Yang et al., 2016).

PEB was measured using the items from Alcock et al. (2020). The scale measures the conscious efforts taken by an individual to conserve the environment and reduce the negative effect of their daily actions on the environment (Alcock et al., 2020). Details regarding the items of the measurement scales are given in Appendix A (Table A1).

Demographic variables such as age, gender, and income were found to strongly influence one's PEB. Therefore, several studies have used them as control variables (Han & Xu, 2020; Luo et al., 2020). Similarly, these variables were controlled during the data analysis for this research, but the effect of these variables on PEB was negligible.

4 | DATA ANALYSIS AND RESULTS

4.1 | Common method variance evaluation

The results of Harman's one-factor test through exploratory factor analysis show that the variance proportion of a single factor is only

TABLE 2 Common latent factor technique.

Items	Standard regression weight without latent factor (1)	Standard regression weight with latent factor (2)	(1)–(2)
SI1	0.879	0.843	0.036
SI2	0.922	0.910	0.012
SI3	0.955	0.837	0.118
SI4	0.809	0.797	0.012
GB1	0.810	0.756	0.054
GB2	0.910	0.911	−0.001
GB3	0.877	0.778	0.099
GB4	0.931	0.899	0.032
GB5	0.911	0.900	0.011
SE1	0.955	0.878	0.077
SE2	0.867	0.853	0.014
SE3	0.925	0.916	0.009
SE4	0.811	0.780	0.031
SE5	0.915	0.890	0.025
CAI1	0.888	0.817	0.071
CAI2	0.933	0.894	0.039
CAI3	0.952	0.867	0.085
PEB1	0.813	0.784	0.029
PEB2	0.933	0.832	0.101
PEB3	0.925	0.910	0.015
PEB4	0.767	0.700	0.067
PEB5	0.911	0.807	0.104
E1	0.912	0.899	0.013
E2	0.920	0.910	0.011
E3	0.745	0.708	0.037
E4	0.911	0.812	0.099
E5	0.876	0.780	0.096
11	0.901	0.876	0.025
12	0.788	0.697	0.091
I3	0.877	0.765	0.112

34%, which is less than the 50% threshold (Podsakoff et al., 2003). Next, the results of the 'Common latent factor analysis' test (Podsakoff et al., 2003), presented in Table 2, show that the difference between the regression weights of the proposed model and the common factor model is <0.2, thus confirming that common method bias is not of much concern.

4.2 | Measurement validity

Prior to the hypotheses testing, the study confirmed the validity and reliability of the measurement items using AMOS 20. Table 3 shows the confirmatory factor analysis results. The overall fit indices are found to be satisfactory (Comparative Fit Index: 0.922; Tucker Lewis Index: 0.910; Root Mean Square Error of Approximation: 0.053; Root Mean Square Residual: 0.023; Normed Fit Index: 0.910; Standardized Root Mean Square Residual: 0.014). These results indicate the appropriateness of the measurement model.

The results, presented in Table 4, confirm the reliability of the constructs. Cronbach's alpha values are greater than the required criteria (0.7) for all the constructs (Hair et al., 2010). Composite reliability (C.R.), factor loadings, and average variance extracted (AVE) were calculated to confirm convergent validity. The factor loadings of all the items and C.R. of all the constructs were greater than the required threshold of 0.7 (Hair et al., 2010). Also, the AVE values were >0.5 and less than C.R. All these results confirm that the model has satisfying convergent validity (see Table 4).

To measure discriminant validity, the following criteria were assessed: as shown in Table 4, the maximum shared variance and average shared variance of the constructs were less than the AVE values. Also, the square root values of AVE were compared with the correlation between the variables. The results, presented in Table 5, show that all the correlation values are less than the square root of the AVE values. These results satisfy the conditions to confirm discriminant validity (Fornell & Larcker, 1981; Hair et al., 2010).

4.3 | Hypotheses testing

The research hypotheses were tested using the SPSS Process Macro. The observed variable approach is used to test the moderated mediation model because it provides specific conditional effects that are more accurate compared with the results provided by other covariance-based approaches (Hayes, 2017). First, the mediation hypotheses were tested using Process Macro Model 4, and then the moderated mediation hypotheses were tested using Process Macro Model 16 (Hayes, 2017). When the model involves *m* independent variables, PROCESS is run *m* times, each time including one variable as the independent variable while treating the remaining $m-1$ independent variables as covariates (Hayes, 2012). Each run of the PROCESS then generates the effects for the variable currently included as the independent variable. We followed the above-mentioned procedure since the study involves three independent variables. The other two variables are kept as co-variates when a particular independent variable is added as the predictor. The process is repeated until each independent variable is included as a predictor and the corresponding moderated mediational results are interpreted.

4.3.1 | Mediation analysis

Following the guidelines of Preacher and Hayes (2008), Process Macro Model 4 was used to analyse the mediation effect of collective intention between GSN affordances and PEB. As indicated in the hypothesized model, the direct effects of green broadcasting and green social endorsement on PEB were significant (0.355, $p < 0.05$; 0.125, $p < .05$, respectively), thus supporting H2 and H3. But the

TABLE 3 Confirmation model results.

Indices	*p* close	CFI	TLI	RMSEA	RMR	NFI	SRMR
Actual	.000	0.922	0.910	0.053	0.023	0.910	0.014
Standard (Hair et al., 2010)	Significant	>0.9	>0.9	<0.08	<0.05	>0.9	Close to 0

Abbreviations: CFI, Comparative Fit Index; NFI, Normed Fit Index; RMR, Root Mean Square Residual; RMSEA, Root Mean Square Error of Approximation; SRMR, Standardized Root Mean Square Residual; TLI, Tucker Lewis Index.

TABLE 4 Factor loadings and validity results.

Items	Factor loadings	Cronbach's alpha	CR	AVE	MSV	ASV
Green self-identification		.89	0.92	0.74	0.09	0.03
SI1	0.87***					
SI2	0.89***					
SI3	0.91***					
SI4	0.78***					
Green broadcasting		.87	0.93	0.72	0.06	0.02
GB1	0.87***					
GB2	0.90***					
GB3	0.81***					
GB4	0.80***					
GB5	0.88***					
Green social endorsement		.91	0.91	0.68	0.08	0.04
SE1	0.89***					
SE2	0.88***					
SE3	0.80***					
SE4	0.79***					
SE5	0.77***					
Collective intention		.92	0.94	0.83	0.11	0.077
CAI1	.93***					
CAI2	.91***					
CAI3	.90***					
PEB		.86	0.90	0.65	0.10	0.07
PEB1	0.79***					
PEB2	0.73***					
PEB3	0.78***					
PEB4	0.89***					
PEB5	0.85***					
Emotional value		.93	0.96	0.82	0.10	0.04
E1	0.91***					
E2	0.93***					
E3	0.92***					
E4	0.90***					
E5	0.89***					
Informational value		.90	0.93	0.81	0.07	0.02
I1	0.92***					
I2	0.87***					
I3	0.93***					

Abbreviations: ASV, average shared variance; AVE, average variance extracted; CR, composite reliability; MSV, maximum shared variance; PEB, pro-environmental behaviour; *** $p < .001$.

TABLE 5 Results of discriminant validity.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Green self-identification (1)	**0.86**	0.01	0.12	0.25	0.30	0.11	0.03
Green broadcasting (2)		**0.85**	0.07	0.23	0.26	0.08	0.09
Green social endorsement (3)			**0.82**	0.28	0.28	0.17	0.07
Collective intention (4)				**0.91**	0.30	0.32	0.28
PEB (5)					**0.80**	0.24	0.23
Emotional value (6)						**0.911**	0.12
Informational value (7)							**0.89**

Note: Diagonal values, written in bold text, show the square root values of AVE. Off-diagonal values show the correlation values among the proposed variables.
Abbreviations: AVE, average variance extracted; PEB, pro-environmental behaviour.

direct effect of green self-identification on PEB is insignificant, as the 95% CI includes 0, thus rejecting H1. Further, the results also confirm that there is a positive influence of GSN affordances (green self-identification, green broadcasting and green social endorsement) on collective intention, as the effects are found to be strongly significant (0.267, $p < .001$; 0.366, $p < .001$; 0.214, $p < .001$), thus confirming H4–H6. Further, as hypothesized (H7), collective intention towards PEB is found to significantly enhance PEB (0.466, $p < .001$).

The effects, both direct and indirect, shall be proven to be statistically significant if zero doesn't straddle between lower-level and upper-level bootstrap confidence intervals (LLCI, ULCI; Hayes, 2017). Following this guideline, it is found that green self-identification has an unconditional indirect effect on PEB through collective intention (0.246, $p < .05$). This result supports H8a. Similarly, the result shows that green broadcasting has an unconditional indirect effect on PEB through collective intention (effect: 0.369; LLCI, ULCI: 0.188, 0.437), thus confirming H8b. Finally, it is also found that green social endorsement has an unconditional indirect effect on PEB through collective intention (effect: 0.108; LLCI, ULCI: 0.065, 0.465), which confirms H8c.

From the direct and indirect effect results, it is confirmed that collective intention has a partial mediating effect on the relationships between green broadcasting and PEB and between green social endorsement and PEB since both the direct and indirect effects are significant. There is a full mediating effect of collective intention between green self-identification and PEB since the direct effect is insignificant. The results are presented in Table 6.

4.3.2 | Conditional indirect effects

Process Macro Model 16 was used to test the moderated mediation hypotheses. This model is suitable for the study because it provides partial moderated mediation indices and facilitates moderated mediation analysis with multiple moderators instead of analysing the model separately for each moderator (Hayes, 2017). Three moderator levels are represented by the mean, one standard deviation above the mean, and one standard deviation below the mean (high, medium, and low, respectively) to examine how the mediating effects act at different levels of the moderator. Bootstrap confidence intervals (CIs) are used to assess the statistical significance of the conditional indirect effects (Hayes, 2017). Also, when the moderated mediation effect is probed for one moderator, the other moderator is conditioned at its mean (Hayes, 2017).

As given in Table 7, the interaction effect of perceived informational value and collective intention on PEB is found to be positive and significant (0.300, $p < .001$). Similarly, the interaction effect of perceived emotional value and collective intention on PEB is found to be positive and significant (0.135, $p < .001$).

Going forward, first, taking green self-identification as the predictor, both moderators were included in the model in the M-Y path. The index of moderated mediation for informational value is non-significant (0.025), 95% CI (−0.001, 0.131). This result demonstrates that, independent of the effect of perceived emotional values, the indirect effect of green self-identification on PEB through collective intention is not moderated by perceived informational values. Therefore, H9a is rejected. However, the index of moderated mediation for perceived emotional value is significant ([0.135], 95% CI [0.006, 0.224]). Further, by looking at the conditional indirect effects, it is understood that the indirect effect of green self-identification on PEB is significant at all levels. However, the effect is stronger at the higher levels of the moderator (0.405, 95% CI: 0.188, 0.599) than at lower and medium levels (0.252, 0.284). This evidence supports H10a, that is, the moderating effect of perceived emotional values is positive and significant on the indirect relation between green self-identification and PEB. The results are illustrated in Table 8.

Next, taking green broadcasting as the predictor, conditional indirect effects were assessed. The results presented in Table 9 show that the partial moderated mediation index is significant for both perceived informational values (0.145, 95% CI: 0.011, 0.231) and perceived emotional values (0.064, 95% CI: 0.006, 0.124). The conditional indirect effect tabulation shows that the indirect effect is significant and greater at higher levels of perceived informational values (0.377) compared with lower (0.011) and medium levels (0.124). This evidence supports H9b. Further, the indirect effect is significant and greater at higher levels of perceived emotional values (0.233), when compared with the effect at lower (0.031) and medium levels (0.105)

TABLE 6 Unconditional mediating effects.

Collective intention					PEB			
Predictor	**Effect**	**SE**	**LLCI**	**ULCI**	**Effect**	**SE**	**LLCI**	**ULCI**
Green self-identification	0.267***	0.039	0.211	0.355	0. 108	0.021	−0.017	0.323
Green broadcasting	0.366***	0.039	0.332	0.572	0. 355*	0.036	0.012	0.455
Green social endorsement	0.214***	0.042	0.027	0.267	0.125*	0.016	0.042	0.176
Collective intention	-	-	-	-	0.466***	0.371	0.381	0.532
Unconditional mediating effects through collective intention								
Predictor	**Effect on PEB**		**SE**		**LLCI**			**ULCI**
Green self-identification	0.246		0.036		0.172			0.356
Green broadcasting	0.369		0.039		0.188			0.437
Green social endorsement	0.108		0.013		0.065			0.465

Abbreviations: LLCI, lower limit confidence interval (95%); PEB, pro-environmental behaviour; SE, standard error; ULCI, upper limit confidence interval (95%).
*p < .05; ***p < .001.

of perceived emotional values. This evidence supports H10b. The results are tabulated in Table 9.

Finally, taking green social endorsement as the predictor, conditional indirect effects were analysed. As depicted in Table 10, the partial moderated mediation index was found to be significant for both perceived informational values (0.025, 95% CI: 0.017, 0.181) and perceived emotional values (0.094, 95% CI: 0.042, 0.234). As given in Table 10, the indirect effect is significant and greater at higher levels of perceived informational values (0.135) when compared with the effects at lower (0.032) and medium levels (0.011). This evidence supports H9c. Further, the indirect effect is significant and greater at higher levels of perceived emotional values (0.266) when compared with the effects at lower (0.006) and medium values (0.156). This supports H10c. The results are presented in Table 10.

5 | DISCUSSION

The empirical evidence from the current research supports most of the hypotheses. It thus confirms the proposed model and generates interesting findings. The significant partial mediating effects of collective intention between two GSN affordances (green broadcasting and green social endorsement) and PEB show that collective intention plays a major role in driving the GSN affordances towards real pro-environmental actions. This is in line with the existing PEB scholarship that demonstrates how we-intentions and shared commitments are necessary to bring behavioural changes that benefit the environment (Biel & Thøgersen, 2007; Smiley, 2017). This very emphasis on collective intentions endorses and extends prominent research studies carried out in the recent past. For instance, in their research on the 'Greta Thunberg Effect', researchers substantiate that being savvy of pro-environmental figures instills collective intentions, which facilitate pro-environmental actions among individuals (Sabherwal et al., 2021). Likewise, underscoring collective intentions in virtual communities, Tsai and Bagozzi (2014) contend that we-intentions can be crucial for gauging altruistic behaviours (such as pro-environmental actions; de Groot & Thøgersen, 2018).

Further, the results show some interesting findings on the strength of mediating effects. Green broadcasting is found to have a stronger direct as well as indirect effect on PEB when compared with that of green self-identification and social endorsement. While scepticism is high towards broadcasted information on social networks (Vraga & Tully, 2021), broadcasted messages through specific social networks may paint a different picture. Specific social networks enhance trust and induce collective intention. That is because trust is formed out of frequent interactions and repeated exposure to a set of people who have shared purposes and interests (Gössling, 2004; Wang et al., 2020). GSNs satisfy the above-mentioned conditions. These reasons make social media users form specific online communities even when they intend to broadcast messages to a larger audience (Potnis & Tahamtan, 2021; Rotman & Preece, 2010). This shows that when individuals receive/send a broadcast message through GSNs, they become members of a larger influencer network. They build social bonding and oneness (Berger & Milkman, 2010), thus leading to PEBs. Besides, communication campaigns, particularly those characterized by the green proposition, can be highly motivating for individuals with aligned interests and induce eco-friendly actions (Jaiswal et al., 2021).

Further, the full mediation finding shows that green self-identification is essential to induce collective intention, which in turn enhances PEB. Collective intention may not always ensure PEB due to the following reasons: While collective intentions are generally constructive, there is a possibility that individuals fall into the normative collective intention due to the need for social belongingness and affective loyalty (Becker & Tausch, 2015). In such cases, collective intention becomes a double-edged sword, as normative collective intention may become a barrier to any social change (Becker & Tausch, 2015). It is therefore essential for the individuals

TABLE 7 Interaction effects.

Interaction effects	Effect	SE	LLCI	ULCI
Collective intention* Informational value	0.300***	0.021	0.008	0.418
Collective intention* Emotional value	0.135***	0.008	0.017	0.411

Abbreviations: LLCI, lower limit confidence interval (95%); SE, standard error; ULCI, upper limit confidence interval (95%); *** $p < .001$.

TABLE 8 Conditional indirect effect of green self-identification on PEB.

Conditional indirect effects				
Predictor	**Level of moderators (when the other moderator is held at constant, i.e., at mean level; Hayes, 2017)**	**Indirect effect on PEB**	**LLCI**	**ULCI**
Green self-identification	−1 SD (INVA as Moderator)	0.113	0.011	0.155
	M (INVA as Moderator)	0.135	0.133	0.145
	+1 SD (INVA as Moderator)	0.163	0.134	0.166
Green self-identification	−1 SD (MVA as Moderator)	0.252	0.076	0.272
	M (MVA as Moderator)	0.284	0.167	0.315
	+1 SD (MVA as Moderator)	0.405	0.188	0.599
Partial Moderated Mediation Index				
Informational values		0.025	−0.001	0.131
Emotional values		0.135	0.006	0.224

Abbreviations: INVA, informational value; LLCI: lower limit confidence interval (95%); MVA, emotional value; PEB, pro-environmental behaviour; ULCI: upper limit confidence interval (95%).

TABLE 9 Conditional indirect effect of green broadcasting on PEB.

Conditional indirect effects				
Predictor	**Level of moderators (when the other moderator is held at constant, i.e., at mean level; Hayes, 2017)**	**Indirect effect on PEB**	**LLCI**	**ULCI**
Green broadcasting	−1 SD (INVA as Moderator)	0.011	0.001	0.100
	M (INVA as Moderator)	0.124	0.012	0.127
	+1 SD (INVA as Moderator)	0.377	0.233	0.599
Green broadcasting	−1 SD (MVA as Moderator)	0.031	−0.144	0.172
	M (MVA as Moderator)	0.105	−0.177	0.115
	+1 SD (MVA as Moderator)	0.233	0.088	0.319
Partial Moderated Mediation Index				
Informational values		0.145	0.011	0.231
Emotional values		0.064	0.006	0.124

Abbreviations: INVA, informational value; LLCI, lower limit confidence interval (95%); MVA: Emotional value; PEB, pro-environmental behaviour; ULCI, upper limit confidence interval (95%).

to first identify themselves with the desired actions. Collective intention driven through such self-identification may reduce the normative aspect and therefore lead to productive behaviour and positive social change. While this finding echoes prior research (for instance, Confente et al. (2020)) in terms of emphasizing green self-identification as a key driver of intentions towards PEBs, it also extends the pertinence of this liaison to GSNs, wherein it has been inadequately explored.

The results of conditional indirect effects show that the indirect effect of green self-identification on PEB is more intense for individuals who perceive high emotional values from the network. Also, it is not affected by individuals with differing levels of perceived informational value (see Table 8). This is in line with previous research that demonstrates the significance of emotional values in strengthening environmentally conscious behaviour (Khan & Mohsin, 2017). The significance of emotional values is observed in GSNs as well (Joshi et al., 2021). Specifically, emotions and affective processes are positively aligned with the concept of self-identity (Konopka et al., 2018; Makkar & Yap, 2018). Social network users react favourably, especially when they perceive emotional appeal from the content (Chen

TABLE 10 Conditional indirect effect of green social endorsement on PEB.

Conditional indirect effects				
Predictor	**Level of moderators (when the other moderator is held at constant, i.e., at mean level; Hayes, 2017)**	**Indirect effect on PEB**	**LLCI**	**ULCI**
Green social endorsement	−1 SD (INVA as Moderator)	0.032	0.004	0.210
	M (INVA as Moderator)	0.011	0.003	0.132
	+1 SD (INVA as Moderator)	0.135	0.017	0.244
Green social endorsement	−1 SD (MVA as Moderator)	0.006	0.004	0.142
	M (MVA as Moderator)	0.156	0.077	0.175
	+1 SD (MVA as Moderator)	0.266	0.188	0.319
Partial Moderated Mediation Index				
Informational values		0.025	0.017	0.181
Emotional values		0.094	0.042	0.234

Abbreviations: INVA, informational value; LLCI, lower limit confidence interval (95%); MVA: emotional value; PEB, pro-environmental behaviour; ULCI, upper limit confidence interval (95%).

et al., 2023; Lee & Hong, 2016). Therefore, individuals with green self-identification may easily convert collective intention into PEB when they perceive high emotional values.

The results also show that both informational and emotional values strongly enhance the existing mediating effect of collective intention between green broadcasting and PEB. This is in line with the SIT, as the theory emphasizes how individuals' attitudes and subsequent behaviours are greatly influenced by the environment, especially when they perceive that the environment is emotionally rewarding and of value (Kelman, 1958). Interestingly, the conditional indirect effects also show that the moderating effect of perceived informational value is relatively greater than that of perceived emotional value on this indirect effect (see Table 9). This inference strongly lays out the significance of informational value in treating the green broadcast message as a strong mechanism for inducing PEB. Studies show that while interacting in online communities, individuals perceiving strong informational value are highly susceptible to behavioural changes; they also experience high satisfaction levels (Hwang & Foote, 2021; Okleshen & Grossbart, 1998; Yang et al., 2016). This is particularly applicable in the context of eliciting environmentally responsible behaviours since valued information enables informed decision-making among individuals (Hosta & Zabkar, 2021). Also, research demonstrates that broadcasted content has an increasingly positive effect on climate action, especially when individuals perceive the information to be credible and worthy (Dong et al., 2018). These arguments are further strengthened by the current research findings. The other significant finding from the analysis is that the indirect effect of green social endorsement on PEB through collective intention is relatively more significant for individuals with high perceived emotional value than for individuals with high perceived informational value (see Table 10). This can be justified by recent research that associates social endorsement cues (e.g., likes) with emotional values (Crapa et al., 2024; Thai & Wang, 2020) and advocates the significance of emotional value in generating real green behaviours.

6 | IMPLICATIONS, LIMITATIONS, AND FUTURE DIRECTIONS

6.1 | Theoretical implications

While researchers have posited that GSNs ameliorate individuals' keenness to contribute towards environmental good (Bhatti et al., 2020; Ghali et al., 2016; Han & Xu, 2020), shaping a positive behavioural inclination towards the environment encompasses much more. Particularly, it entails catering to the deep-rooted psychological intricacies with specific GSN affordances as the frame of reference. This study, in this regard, facilitates a strong foundation for the green IS literature. Unlike traditional celebrity endorsement, social endorsement rooted in GSNs is a voluntary action performed by peers. It is devoid of monetary incentives for the endorsers (Thai & Wang, 2020), thus makes social endorsement more effective in facilitating PEBs through GSNs. Similarly, green broadcasting not only entails communication with a larger audience but also ensures reaching the right audience. Finally, green self-identification, though highlighted in the PEB literature, has not been investigated as a GSN affordance that initiates PEBs. In essence, by weaving these affordances alongside PEB, this study provides a strong reason for researchers to focus on GSN affordances to facilitate PEB. It thus resonates with and responds to existing works that contend that social networks can promote PEB, and dwelling into the nitty-gritty of the process will offer valuable insights for theory and practice (Shen et al., 2023; Tian & Liu, 2022).

Further, the existing literature on social network–PEB relationships has largely adopted a silos approach to the application of theory (Tian & Liu, 2022). Therefore, the conjugation of the SIT and the S-O-R framework to explain the *GSN affordances-collective intention–PEB* relation represents a key theoretical advancement to the existing IS-PEB literature. As per scholars, such conjugation should be driven by certain considerations, such as the *proximity* in basic assumptions underlying the theories/frameworks (Okhuysen & Bonardi, 2011). Proximity denotes the 'conceptual distance that exists between the phenomena that the lenses address in their original conception' (Okhuysen & Bonardi, 2011, p. 7). That said, the SIT and the S-O-R framework both somewhat pertain to elucidating individual actions/behaviours as outcomes. Furthermore, both have a psychological aspect as a key component innate in individual decision-making with regards to manifesting a certain behaviour. Jacoby (2002), in the S-O-R framework, put forward that the psychological character intrinsic to the stimulus guides the organism phase and the individual's behavioural response. Likewise, in the SIT, scholars (Kelman, 1958; Spreng & Page Jr, 2001) argue that the internalization of social influence exerted by a certain community entails psychological processing of the resources furnished by the community; this influences one's beliefs and shapes subsequent behaviour. Thus, SIT and S-O-R are proximal. Weaving them together refines the way we comprehend how and why individuals exhibit certain behaviour in varied contexts. Furthermore, behavioural scholarship affirms that the consolidation of individual and social factors for determining individual behaviour offers avenues for original and ingenious contributions (Ostroff, 1993). This study therefore blends the SIT and the S-O-R framework to introduce a socio-psychological perspective. This way, it contributes towards fortifying our grasp of the phenomenon entailing GSN affordances, collective intention, and PEBs.

Further, with regards to the mediating effect of collective intention in the *GSN affordances–PEB* relationship, the study reckons that collective intention can be a process lending effectiveness to the liaison between GSN affordances and environmental behaviours. While some of the affordances may not accrue desirable results, this study explains how collective intentions are essential to accruing desirable behaviour from the potentialities of GSN. For example, the findings illustrating the full mediation of collective intention between green self-identification and PEB offer a theoretical explanation of how important collective intentions are in order to treat green self-identification as a motivator of PEB. This extends existing works that stress the importance of self-identification for garnering PEBs (Becerra et al., 2023; Wallace & Buil, 2023). Additionally, this broadens the scope of collective intentions, as recommended by certain research and review studies (Landmann & Rohmann, 2020; Masson & Fritsche, 2021).

The results also indicate that GSN affordances exert a stronger influence on individual PEB through collective intention amid high levels of perceived informational and emotional values. That is, the coherence between collective intentions and perceived values is crucial to prompting PEB among GSN users. And so, within the GSN, investing in efforts appealing to individuals' cognitive and affective states is expected to harness PEB among the network members. Grounded on the SIT and the S-O-R, the moderated mediation findings: (i) serve to alleviate ambiguities pertaining to the role of values, not only in the context of PEBs (Grimmer & Miles, 2017), but also in respect of social networks' behavioural impact (Kwon & Namkung, 2022; Shen et al., 2023), and (ii) significantly augment related research on the subject (Hassan et al., 2016; Pichert & Katsikopoulos, 2008; Yang et al., 2016).

The study also enriches the literature on the S-O-R framework by illustrating how organisms sometimes interact with each other to produce a desirable response (Jacoby, 2002). By introducing and investigating how perceived information and emotional values interact with collective intention, this study becomes one of the first to indicate that the perceived values offered by the network condition the way in which GSN affordances emerge as facilitators for PEBs through collective intentions. It thus expands the research on social networks too by putting forward ways to ameliorate user experiences and the effectiveness of social networks, specifically GSNs.

6.2 | Practical implications

The research fundamentally provides strong implications for environmentalists and environmental organizations by helping them identify effective mechanisms to influence online community members to engage in environmental actions. Environmental activists (both individuals and firms) can revisit the action possibilities of social networks and influence users to use/receive those possibilities.

The identification of the roles of perceived informational and emotional values will help activists and application designers strategize the affordances accordingly. For example, online community designers can develop innovative digital cues to facilitate social endorsement among the network members, which may not only reduce the cost of acquiring a celebrity for endorsement but may also facilitate social belongingness and pro-environmental actions. The significant full mediating effect of green self-identification indicates that environmentalists should post content (e.g., a video containing how one can reuse their clothes and create a trend while greatly saving the environment) in such a way that the users may identify themselves with the content and further connect 'I' with 'We' to transform their collective intentions into actions.

Further, to understand why certain environmental online communities fail to create an impact on individuals' PEB, environmentalists first need to understand when and how the GSN affordances affect individuals' intentions and actions. The findings show that collective intention works best as a mediating facilitator between green self-identification and PEB, especially when perceived emotional values are high. This indicates that both, environmentalists and green online community application developers, should understand the significance of emotional content. They can strategize the network's communication with emotional content so that individuals will have stronger motivation to convert the green self-identification into PEB through collective intention. This research further enriches the existing knowledge on the social network–PEB link by indicating the significance of perceived informational value in strengthening the link. These implications make it necessary for environmentalists to develop strategies that may address the gap between intentions and actions, which is very prevalent when employing social networks to induce environment-related actions.

6.2.1 | Implications with regards to the post-pandemic state of affairs

It has been 3 years since the outbreak of the COVID-19 pandemic. The pandemic scourged human lives in countries the world over and caused economic and social disruption. But at the same time, it proffered an opportunity to recover in an enhanced way and strive for sustainable growth. Crucially, endeavours to realize the Sustainable Development Goals have scaled like never before. Efforts are being made not only by governments, industry leaders, and appropriate agencies but also by the common people/interest groups to address environmental degradation. An important thing to note is that, though devastating in nature, the pandemic has ushered in a *tech revolution* (Gurnani, 2022). Consequently, adopting the 'new normal', these various interest groups are all embracing the digital route to shaping a better post-pandemic world. Herein, the current study and its findings hold significant implications. First, it offers a comprehensive process that the concerned stakeholders can use to leverage social networks to encourage PEB. It thus offers an optimal way to channelize digital technology to the intended end. As indicated by the study model, earnest efforts would entail a focus on enhancing the utility of GSN affordances of green self-identification, green broadcasting, and green social endorsement. And in line with the S-O-R, nurturing collective intention would very well serve to generate PEB. As illustrated by prior research, getting people to converse about green issues in an evocative manner such that these talks effectuate behaviour can be quite surmountable. With collective intention as the mediating variable, environmentalists can initiate virtual meetups and live-streaming sessions that can promote further interactions. Doing so will lead to a sense of collective as well as individual responsibility. These initiatives will prompt individuals on personal as well as larger levels to engage in pro-environmental actions. In consonance, a handful of prior studies describe these intentions as pertinent and suggest utilizing them in virtual communities for attaining positive behavioural responses (Harland et al., 2007). Additionally, the findings show that when perceived informational and emotional values are high, GSN affordances will have a high impact on PEB through collective intention. For instance, an impactful idiosyncrasy of a GSN is to reinforce environmentally conscious policies. Though everyone in the community wants to abide by these, the entire purpose may become diluted or overlooked amidst the several other interactions. Therefore, it is necessary to strategize communication that entails high emotional value (e.g., usage of emoticons and affective language evoking high-arousal emotions) and informational value (e.g., usage of statistics, sharing information sources, evidence, updates, and other useful content). Through these strategies, practitioners can effectively use affordances like green self-identification, broadcasting mechanisms (e.g., Live-stream, content sharing), and social endorsement mechanisms (such as paralinguistic digital affordances [likes, comments, and repost content]) to remind and encourage members to follow environmental policies (e.g., policies pertaining to recycling, conscious consumption, etc.). In so doing, the GSNs can be better set to foster pro-environmental activities and achieve their goals.

6.3 | Limitations and future directions

While the current study ensures a time-lagged three-wave design, future researchers could validate the causal inferences by introducing multiple sources during data collection. Also, providing a time lag between the data collection of mediating and moderating variables is suggested. By measuring collective intentions at time 2, and PEB at time 3, this study provides a temporal separation between the assessment of collective intentions and actual behaviours. However, stronger longitudinal studies may validate the relationship between collective intentions (futuristic) and PEBs (actual).

Further, this study focuses on how informational and emotional values play a role in accentuating the effect of collective intention on PEB. However, the study did not focus on how social networks can be used to induce these values or what makes individuals perceive these values in GSNs. The literature on PEB communication would benefit future research by conducting an intervention study on the effect of different communication forms on informational and emotional values among GSN users. Also, in addition to the proposed moderators, we encourage future researchers to capture specific contextual and individual dimensions. For instance, cultural factors, individuals' behavioural spill-over effects, and personality traits may impact PEB (Agag et al., 2020; Esfandiar et al., 2022). This exploration may help researchers further narrow down the intention–behaviour gap.

Future research can also perform a comparative investigation between online and offline PEB communities and behaviours. For example, the investigation on how exclusive online networks enhance PEBs within an online environment, such as signing petitions on environmental websites (Feitosa & Gozetto, 2018), and outside the online environment, such as green purchases (Luo et al., 2020), would provide an interesting contribution to the PEB research. While this study provides a novel understanding of the perceived social influence on individualistic behaviours by investigating the collective intention–PEB relationship, we call upon future researchers to investigate the comparative effects of individual intentions and collective intentions on online and offline PEBs. Also, recent research studies have claimed that individuals may indulge in conspicuous pro-environmental online and offline behaviours to predominantly seek validation (Wallace & Buil, 2023). However, considering the varying degrees of conspicuousness and authenticity across online and offline PEBs (Wallace & Buil, 2023), it is essential for future researchers to investigate both the benefits and pitfalls of GSN-induced online and offline PEBs.

This research focussed on the users of all forms of GSNs because online communities are diverse in nature and spread across several online platforms (Malinen, 2015). However, GSNs created and managed by different stakeholders (e.g., regional NGOs, global bodies like the UN, etc.) may generate different affordances and different effects on the users' PEBs. Future researchers can investigate these differential effects to better understand how GSN affordances facilitate PEBs for different stakeholders.

Finally, the current research utilizes a convergence of the SIT and the S-O-R framework to explicate the relationship between GSNs and PEB. However, as suggested by recent review studies (for instance, Lin, Xu, & Wang, 2022; Lin, Zhu, et al., 2022; Paul & Bhukya, 2021), diverse theoretical approaches (e.g., the prospect theory and the technology adoption model) hold pertinence in the PEB domain, and so utilizing these will pave the way for further insights into the GSNs–PEB relationship. Also, given that a singular approach to the application of theory has been predominantly utilized by prior research (Lin, Xu, & Wang, 2022; Lin, Zhu, et al., 2022; Xiao et al., 2023), the integrated theoretical lens utilized by this study will serve as a guiding platform for future enquiry into the subject.

7 | CONCLUSION

Heightened interest in ways in which technology can be utilized to steer PEBs, post-covid, has accrued into burgeoning research endeavours into social networks–PEB interaction. In this regard, this study posits GSNs as an apt tool. It provides empirical evidence of the critical factors that would make GSNs successful and evinces the need to place GSNs as the practitioners' focal considerations in their efforts aimed at evoking PEBs. As discussed, specific GSN affordances will enhance social network users' PEB via collective intention when they perceive the content to be of high informational and emotional value. That is, interventions addressing the users' informational and emotional requirements will lend credence to the use of GSNs for bringing about their PEB. The study, with its theoretical approach and significant implications, broadens the spectrum of existing social network-PEB scholarship and offers recommendations of import to theorists and practitioners. All in all, social networks and exclusive online communities are here to integrate several other IS applications and individuals' behaviours, creating a never-before-seen social impact. Therefore, we call on future researchers to enhance the proposed model to further expand knowledge on the GSN–PEB relationship.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ENDNOTES

[1] Online community is defined as follows: 'An online community consists of people interacting socially and sharing a purpose, of policies to guide these interactions, and of computer systems to facilitate the sense of togetherness' (Preece, 2000, p. 10). Examples include online communities designed specifically for brand-related information sharing (Zhou et al., 2020), environment-based discussions (Rokka & Moisander, 2009), and discussions centred on problems faced by ethnic communities (Klassen et al., 2021).

[2] GSNs are created by environmental stakeholders to facilitate user-generated, continuous, and dynamic environmental dialogue (e.g., sharing videos and pictures related to environmental activities) and promote PEB (Ghali et al., 2016).

[3] Vertical networks such as Care2 (www.care2.com) are niche social network platforms centred around specific interests for like-minded people to engage in a more private and focussed setting.

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APPENDIX A

TABLE A1 Annexure measurement.

S. no.	Items (think about the green social networks [GSN] you are using or have used when answering these questions)	Construct and source
E1	It is a proud feeling to be a part of GSN	Emotional value (Joshi et al., 2021; Sweeney & Soutar, 2001)
E2	The way GSN makes me feel motivates me to use the GSN often	
E3	The usage of GSN makes me feel refreshed	
E4	GSN makes me feel good when I use it	
E5	GSN gives me pleasure	
I1	I receive relevant information when I use GSN	Informational value (Okleshen & Grossbart, 1998; Yang et al., 2016)
I2	GSN is very helpful for me to gather sufficient information related to environment	
I3	The information provided in GSN saves me a lot of time	
PEB1	I am able to recycle the waste effectively	Pro-environmental behaviour (Alcock et al., 2020)
PEB2	I buy eco-friendly products	
PEB3	I choose to walk or ride a cycle whenever possible	
PEB4	I often volunteer for environment related activities	
PEB5	When I meet someone face to face, I persuade them to indulge in conservation related to environment activities	
CAI1	I along with my social networking members, intend to sign a petition to conserve the environment when necessary	Collective intention (Iyer et al., 2007; Shen et al., 2013)
CAI2	We (I and the members of the GSN) intend to actively engage in pro-environmental behaviour in the future	
CAI3	We (I and the members of the GSN) plan to use the GSN to act together in resolving environmental-related issues	
SI1	The GSN offers the opportunity to identify my purpose for the environment	Green self-identification (Harrigan et al., 2018)
SI2	GSN helps me identify myself with the people in GSN	
SI3	GSN facilitates personal connection between me and environment	
SI4	GSN helps me become the kind of environmentally responsible person I want to be	
GB1	GSN allows me to share posts related to green activities with all the members of the network	Green broadcasting (Thai & Wang, 2020)
GB2	GSN allows me to initiate posts related to green activities in the network	
GB3	GSN allows me to share pictures, videos and images related to green activities with all the members of the network	
GB4	GSN facilitates the timely delivery of announcements related to environmental activism to all the members of the network	
GB5	GSN allows me to post comments on others' posts that are available for all the members to see	
SE1	GSN facilitates users to voice their opinion through recommendation buttons	Green social endorsement (Gan, 2017; Thai & Wang, 2020)
SE2	GSN facilitates individuals to like a content displayed on the network	
SE3	GSN facilitates the visibility of number of positive reactions of the members about a particular content	
SE4	Through positive emojis, GSN influences one's view of the content	
SE5	GSN allows me to click 'like' when I see a huge number of likes for a particular comment on environment-related activities	

Note: The affordance research also suggests that perceived affordances may lead to behavioural outcomes for both actors and non-actors in several significant ways (Leidner et al., 2018). Similarly, this study claims that the affordances may affect both active and non-active users in the GSN. For instance, even if the members of the GSN do not actively participate in green broadcasting, they might still benefit from the affordances as they may act as receivers of the broadcasted messages. When individuals perceive that the GSN affords voice behaviours through recommendation buttons, it is also understood that they are interacting with the system or the actors in the system. Therefore, even if the individuals are not actively participating in the system, they can be passive actors and may indulge in passive consumption such as receiving, reading and observing broadcasted messages, posts and comments (Lee et al., 2019). Users can be engaged as well as benefit even through non-active behaviours, such as non-clicking on a comment (Ellison et al., 2020; Koh & Kim, 2004). In sum, both active (e.g., creating content) and passive usage (viewing the content/observing the members and their posts) often co-occur in real life, and therefore, both should be considered in social network-based research (Ellison et al., 2020; Koh & Kim, 2004). Since social network engagement is often considered a continuum form of engagement (Alsaad et al., 2023), the measurement items for the current research were adapted, considering both active contributors and passive consumers.